

Natural-Tile Geometry and Cage Accessibility Specification

Stage 11: Exact Reference Tiles, Topological Windows, Conservative Probe Witnesses, and Periodic Accessibility Networks

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Purpose and stage boundary

Stage 11 begins the downstream physical interpretation branch after a certified periodic natural tiling:

```
accepted natural tiling
-> exact reference tile geometry
-> topological windows and translation-labelled adjacency
-> explicit cage/portal probe witnesses
-> periodic accessible-network rank
```

Runtime/API target:

mdstats 0.19.28a0

Primary modules:

mdstats/analysis/tiling_geometry.py
mdstats/analysis/cage.py

LTA integration helper:

mdstats/analysis/lta_natural_tiling.py
build_lta_natural_tiling_reference(...)

This first backend is deliberately bounded. It realizes the exact source-bound reference embedding and requires strictly convex planar faces and convex tiles. It certifies accessibility only at explicit witnesses under periodic spherical obstacles. A blocked witness is **not** promoted to global inaccessibility.

Scientific distinctions

The implementation preserves four separate concepts.

Natural tile

A natural tile is a persistent topological 3-cell of the accepted periodic cell complex. Its identity is determined by scientific faces and translation-labelled attaching maps. Geometry does not redefine that identity.

Topological window

A topological window is one scientific face orbit shared by exactly two translated tile sides. It exists independently of guest size, atom radii, or instantaneous aperture.

Accessible cage

A natural tile becomes an accessible cage for a probe only when an explicit interior witness clears every admitted periodic obstacle sphere by at least the probe radius.

Accessible portal

A topological window becomes an accessible portal only when one explicit in-plane aperture witness:

1. contains a disk of the requested probe radius inside the convex polygon;
2. clears all periodic obstacle spheres by at least the probe radius; and
3. connects two tile placements whose cage witnesses are accessible.

These promotions are probe- and obstacle-model dependent. They never alter the scientific natural tiling.

External methods and original construction

The translation-labelled quotient convention follows Chung, Hahn, and Klee [1]. Natural-tile and properness semantics follow Blatov, Delgado-Friedrichs, O’Keeffe, and Proserpio [2]. The stored sphericity descriptor is Wadell’s true sphericity, the equal-volume-sphere surface area divided by the tile surface area [3].

The following mechanisms are project-specific mdstats constructions:

- exact lifted reconstruction of every tile-side polygon from scientific face placement plus attaching translation;
- exact convex supporting-halfspace validation against a rational interior witness;
- exact tetrahedral fan volume and first-moment accumulation in fractional coordinates;
- topological-window reconstruction directly from the two incidences of each scientific face orbit;
- complete nearest-periodic-image enumeration for a point witness using a metric-derived finite integer range, not a fixed image shell;
- conservative accessibility states that distinguish a sufficient witness from an unresolved blocked witness; and
- translation-rank classification of the accessible quotient network.

No external global pore-finding, Voronoi-network, or maximum-inscribed-sphere algorithm is claimed or silently approximated.

Inputs and source binding

build_tiling_geometry_catalog

```
build_tiling_geometry_catalog(  
    complex_: PeriodicCellComplex,  
    embedding: PeriodicNetEmbedding,  
    ring_index: PrimitiveRingIndex,  
    *,  
    resources: TilingGeometryResources | None = None,  
) -> TilingGeometryCatalog
```

Required source equalities are

```
complex_.periodic_net_embedding_digest == embedding.digest  
complex_.primitive_ring_catalog_digest == ring_index.catalog_digest  
complex_.topology_graph_digest == ring_index.topology_graph_digest
```

The scientific complex must already satisfy the Stage-9 chain, shell, and face-incidence invariants. The first geometry backend additionally requires:

- at least three vertices per face;
- exact planarity of each face polygon;
- strict polygon convexity;
- a convex tile supported by every oriented face plane;
- a strictly interior rational witness; and
- positive exact tile volume.

assess_cage_accessibility

```
assess_cage_accessibility(  
    geometry: TilingGeometryCatalog,  
    embedding: PeriodicNetEmbedding,  
    probe: AccessibilityProbe,  
    obstacles: Sequence[PeriodicObstacleSphere] = (),  
    *,  
    resources: CageAccessibilityResources | None = None,  
) -> CageAccessibilityCatalog
```

All lengths use the Cartesian metric of the unit-volume reference embedding. Obstacle coordinates are fractional and periodic. Obstacle and probe radii must use the same Cartesian length unit.

Persistent data model

Tile geometry

```
NaturalTileGeometry(  
    tile_index: int,  
    label: str,  
    vertex_count: int,  
    edge_count: int,  
    face_count: int,  
    side_indices: tuple[int, ...],  
    fractional_center: tuple[Fraction, Fraction, Fraction],  
    cartesian_center: tuple[float, float, float],  
    fractional_volume: Fraction,  
    cartesian_volume: float,  
    surface_area: float,  
    equivalent_sphere_radius: float,  
    sphericity: float,  
    diameter: float,  
    convex_certified: bool,  
)
```

`fractional_center` is the exact volume centroid of the convex tile, not merely the arithmetic mean of vertices.

Oriented tile sides

```
TileSideRef(  
    tile_index: int,  
    face_index: int,  
    face_image_shift: LatticeShift,  
    incidence_orientation: Literal[-1, 1],  
)  
  
TileFaceGeometry(  
    side_index: int,  
    side: TileSideRef,  
    ring_size: int,  
    fractional_vertices: tuple[RationalPoint, ...],  
    fractional_center: RationalPoint,  
    cartesian_center: tuple[float, float, float],  
    outward_unit_normal: tuple[float, float, float],  
    area: float,  
    perimeter: float,  
    aperture_witness_radius: float,  
)
```

The face center is the equal-weight polygon-vertex average. For a strictly convex polygon this point lies in the polygon interior and supplies a deterministic aperture witness.

Topological windows and adjacency

```
TopologicalWindow(  
    window_index: int,  
    face_index: int,  
    face_digest: str,  
    ring_size: int,  
    side_a: TileSideRef,  
    side_b: TileSideRef,  
    relative_tile_translation: LatticeShift,  
    self_adjacent: bool,  
    area: float,  
    aperture_witness_radius: float,  
    fractional_center: RationalPoint,  
    cartesian_center: tuple[float, float, float],  
)  
  
TileAdjacencyArc(  
    arc_index: int,  
    window_index: int,  
    source_tile_index: int,  
    target_tile_index: int,  
    target_image_shift: LatticeShift,  
)
```

Each window produces two reverse directed arcs. Self-image adjacency is retained as a tile orbit connected to a nonzero translated image of itself.

Probe and obstacle model

```
AccessibilityProbe(radius: float, label: str = "probe")  
  
PeriodicObstacleSphere(  
    obstacle_id: int,  
    fractional_coordinate: RationalPoint,  
    radius: float,  
    label: str = "",  
)
```

The obstacle model is explicit evidence. The module does not infer van der Waals, ionic, or framework radii from chemical species.

Accessibility states

WitnessAccessibilityStatus.CERTIFIED_ACCESSIBLE_AT_WITNESS
 WitnessAccessibilityStatus.WITNESS_BLOCKED_UNRESOLVED

The second state means only that the stored deterministic witness failed. Another point, a nonspherical path, or a more complete geometric search may still prove accessibility.

Exact tile realization

For one tile orbit, each attaching term supplies a translated face placement. The exact lifted polygon is

$$P_{f,\mathbf{t}} = \{\mathbf{x}_i + \mathbf{t}\}_{i=0}^{m-1}.$$

The union of exact polygon vertices supplies a provisional rational interior point

$$\mathbf{c}_0 = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i.$$

For a convex full-dimensional polytope, a positive average over all vertices lies in its interior. Each face plane is oriented so that

$$\mathbf{n}_f \cdot (\mathbf{c}_0 - \mathbf{p}_f) < 0.$$

The backend then requires every tile vertex to satisfy

$$\mathbf{n}_f \cdot (\mathbf{v} - \mathbf{p}_f) \leq 0$$

for every face. All tests use exact rational signs in fractional coordinates.

Volume and centroid

Each outward-oriented convex face polygon is triangulated as a boundary fan. For one triangle $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ and interior point $\mathbf{c}_0$, its tetrahedral volume is

$$V_t = \frac{1}{6} |\det(\mathbf{a} - \mathbf{c}_0, \mathbf{b} - \mathbf{c}_0, \mathbf{c} - \mathbf{c}_0)|.$$

The tile volume is

$$V = \sum_t V_t,$$

and the exact first moment is

$$V\mathbf{c} = \sum_t V_t \frac{\mathbf{c}_0 + \mathbf{a}_t + \mathbf{b}_t + \mathbf{c}_t}{4}.$$

The reference embedding has unit Cartesian cell volume, so exact fractional tile volume equals Cartesian volume numerically.

Surface descriptors

For Cartesian volume V and surface area A , the equal-volume sphere radius is

$$r_{\text{eq}} = \left(\frac{3V}{4\pi} \right)^{1/3}.$$

Wadell sphericity is

$$\Psi = \frac{\pi^{1/3} (6V)^{2/3}}{A}.$$

The diameter is the largest Cartesian distance between tile vertices. These are derived descriptors; they do not enter tile identity or natural-tiling certification.

Window translation and adjacency

Suppose the two tile representatives contain face f at attaching shifts $\mathbf{s}_a$ and $\mathbf{s}_b$. Aligning the same physical face requires the second tile image

$$\mathbf{T}_{a \rightarrow b} = \mathbf{s}_a - \mathbf{s}_b.$$

The reverse arc stores $-\mathbf{T}_{a \rightarrow b}$. A self-adjacent tile orbit with nonzero $\mathbf{T}$ is therefore represented without collapsing periodic connectivity.

Aperture witness

For one strictly convex face polygon, let $\mathbf{q}$ be the Cartesian image of the equal-weight vertex average. The stored geometric witness radius is

$$r_{\text{poly}}(\mathbf{q}) = \min_{e \in \partial P} \text{dist}(\mathbf{q}, e).$$

This certifies that a planar disk of radius r_{poly} centered at $\mathbf{q}$ is contained in the polygon. It is not claimed to be the maximum inscribed circle unless symmetry makes it so.

Complete nearest periodic obstacle search

For witness fractional coordinate $\mathbf{q}$, obstacle coordinate $\mathbf{x}$, integer image $\mathbf{n}$, and row-cell matrix H , the center distance is

$$d(\mathbf{n}) = \| (\mathbf{q} - \mathbf{x} + \mathbf{n})H \|_2.$$

A componentwise rounded image supplies an initial finite upper bound R . Since

$$\mathbf{y} = (\mathbf{q} - \mathbf{x} + \mathbf{n}) = ((\mathbf{q} - \mathbf{x} + \mathbf{n})H)H^{-1},$$

Cauchy–Schwarz gives the finite component bounds

$$|y_i| \leq R \| (H^{-1})_{:,i} \|_2.$$

The backend enumerates every integer image in those metric-derived ranges and therefore does not assume a fixed $3 \times 3 \times 3$ image shell.

Obstacle-surface clearance is

$$\delta = \min_j (d_j - r_j).$$

A probe of radius r_p is certified at the witness when

$$\delta \geq r_p.$$

For portals the polygon witness also requires

$$r_{\text{poly}} \geq r_p.$$

Accessible-network rank

Only arcs whose portal witness and both endpoint cage witnesses are certified are retained. A spanning forest assigns image potentials $\mathbf{p}_i$ to tile orbits. Every non-tree arc $i \rightarrow j$ with translation $\mathbf{t}_{ij}$ produces a cycle voltage

$$\mathbf{z} = \mathbf{p}_i + \mathbf{t}_{ij} - \mathbf{p}_j.$$

The rational rank of the nonzero voltage vectors classifies the periodic accessible component:

Rank	Interpretation
0	isolated cage
1	one-dimensional channel
2	two-dimensional layer
3	three-dimensional network

This is a topological periodic-rank statement for the certified witness graph, not a diffusion-rate or energetic-transport prediction.

LTA integration

`build_lta_natural_tiling_reference(topology)` reruns the exact Stage-10D $K = 8$ construction and returns transient source objects:

```
LtaNaturalTilingReference(
    view,
    discovery,
    embedding,
    ring_index,
    complex,
    partition,
)
```

It creates no second persistent tiling identity. The returned complex and ring index can be passed directly to `build_tiling_geometry_catalog`.

The real-LTA gate must recover:

```
10 tiles
58 topological windows
116 directed adjacency arcs
exact total fractional volume = 1
6 [4^6] tiles
2 [4^6.6^8] tiles
2 [4^12.6^8.8^6] tiles
```

Resource limits

Geometry

```
TilingGeometryResources(
    max_tiles=4096,
    max_faces=16384,
    max_vertices_per_face=128,
    max_pair_distance_tests=5_000_000,
)
```

Accessibility

```
CageAccessibilityResources(
    max_obstacles=100_000,
    max_periodic_image_tests=5_000_000,
    max_network_arcs=1_000_000,
)
```

Resource failures occur before the declared bound is exceeded and never return a partial persistent catalog.

Serialization and reproducibility

Both catalogs use deterministic canonical JSON and SHA-256 digests. Persistent loading is source replay:

```

TilingGeometryCatalog.from_dict(
    payload,
    complex=complex_,
    embedding=embedding,
    ring_index=ring_index,
)
CageAccessibilityCatalog.from_dict(
    payload,
    geometry=geometry,
    embedding=embedding,
)

```

A payload with a recomputed outer digest is still rejected when any geometric or accessibility field disagrees with deterministic reconstruction.

Edge cases and failure policy

The first backend rejects or leaves unresolved:

- nonplanar, degenerate, or non-strictly-convex scientific faces;
- nonconvex tile shells;
- source-digest mismatches;
- inconsistent translated face areas or aperture witnesses;
- more or fewer than two tile sides per scientific face;
- malformed obstacle IDs or nonfinite radii;
- resource exhaustion;
- blocked cage or portal witnesses, which remain unresolved globally;
- frame-dependent chemical geometry not represented by the reference embedding;
- nonspherical guests or obstacles;
- flexible-window path search;
- maximum-inscribed-sphere or full pore-network optimization;
- automatic radii assignment from species; and
- transport kinetics or free-energy barriers.

Focused validation requirements

The Stage-11 focused suite must cover:

1. exact unit-cube volume, centroid, surface area, edges, and diameter;
2. self-image topological windows and reverse translation arcs;
3. rank-three obstacle-free accessible cube network;
4. a probe larger than the stored portal witness;
5. periodic obstacle images at a cell boundary;
6. a blocked cage witness without a false global-inaccessibility claim;
7. transactional resource preflight;
8. deterministic replay and tamper rejection;
9. source-type rejection; and
10. the real-LTA 10-tile, 58-window, unit-volume ground gate.

Deferred extensions

The next implementation rounds may add:

- geometry mapped onto arbitrary compatible trajectory or ensemble frames;
- explicit linker-atom and van der Waals surfaces;
- automatic chemistry-aware radius policies;
- maximum-clearance points and flexible portal paths;
- nonconvex tile decomposition;

- guest-specific connected free-volume regions;
- cage occupancy and portal-crossing events over trajectories; and
- coupling to diffusion and residence-time statistics.

References

- [1] S. J. Chung, T. Hahn, and W. E. Klee, “N-dimensional crystallography,” *Acta Crystallographica A* **40**, 42–50 (1984), doi:10.1107/S010876738400010X.
- [2] V. A. Blatov, O. Delgado-Friedrichs, M. O’Keeffe, and D. M. Proserpio, “Three-periodic nets and tilings: natural tilings for nets,” *Acta Crystallographica A* **63**, 418–425 (2007), doi:10.1107/S0108767307038287.
- [3] H. Wadell, “Volume, shape, and roundness of quartz particles,” *The Journal of Geology* **43**, 250–280 (1935), doi:10.1086/624298.